

26th CIRP Design Conference

Modular Platform Design for Mechatronic Systems using Axiomatic Design and Mechatronic Function Modules

Guenther Schuh, Stefan Rudolf, Stefan Breunig*

*Laboratory for Machine Tools and Production Engineering WZL at RWTH Aachen University, Steinbachstr. 19, 52074 Aachen, Germany** Corresponding author. Tel.: +49-241-80-28196; fax: +49-241-80-628196. E-mail address: S.Breunig@wzl.rwth-aachen.de**Abstract**

This paper introduces a method for the design of modular product platforms (MPP) for highly integrated mechatronic products applying axiomatic design and a system decomposition into mechatronic function modules (MFM). In the first step, using an extended axiomatic design approach, the complex system is decomposed into MFM. The application of axiomatic design ensures a strong function-orientation which is required in mechatronic system development. In the second step, the MFM are designed for use in a product platform. The methodology is applied in a case study considering the design of a MPP for electromechanical steering systems for use in personal vehicles.

© 2016 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Peer-review under responsibility of the organizing committee of the 26th CIRP Design Conference

Keywords: modular product platform; platform design; mechatronic systems; axiomatic design; mechatronic function modules

1. Introduction

When designing mainly mechanical systems the development strongly focusses on physical components and their dependencies. The design of mechatronic systems requires a stronger functional orientation [1]. Functions are realized by elements like sensors, actuators, microcontrollers and mechanical parts which show strong interdependencies [2]. A function-oriented approach further enables the integration of the different involved disciplines: mechanics, electronics/ electronics and software.

Further today's companies try to satisfy increasingly individual customer requirements in saturated markets through and increasing number of product variants. Diverse market requirements and high variety lead to an increased complexity of products and organizations [3]. To manage this complexity modular product platforms have proven to be an useful approach [4]. Current approaches for the design of modular product platforms however often lack a function orientation which is needed for the design of mechatronic systems [5].

In order to address these challenges this paper proposes an approach for the design of modular product platforms for

mechatronic systems. The strong function orientation will be ensured by the use of axiomatic design and mechatronic function modules.

2. Current state of research

In the following the current state of research regarding the design of mechatronic systems and the design of modular product platforms is presented. Furthermore the basics of axiomatic design are described.

2.1. Design of mechatronic systems

This paper addresses the challenge of designing modular product platforms for mechatronic systems. Therefore the relevant basics of mechatronics will be highlighted. According to BUUR mechatronics can be described as “technology which combines mechanics with electronics and information technology to form both functional interaction and spatial integration in components, modules, products and systems” [2]. Especially the functional interaction and the high degree of integration are reasons for an increased

complexity of these systems. The development of mechatronic systems is an interdisciplinary process which has to regard different system levels. An approach for the design of such systems is provided by the V-model [6]. The V-model is a cycle which describes the system design starting with requirements on system level, followed by a discipline specific design and a subsequent system integration. For the structuring of a mechatronic system LUECKEL ET AL. suggest a system decomposition into mechatronic function modules (MFM). An MFM is defined as the basic element of a mechatronic system which is needed to fulfill a specific function. It consists of “sensors, actuators, information processing elements and a related mechanical structure” [7]. The generic structure of a MFM is shown in figure 1.

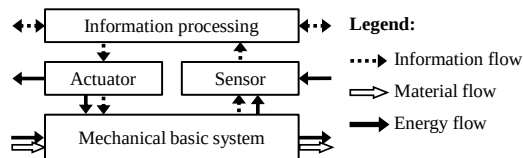


Fig. 1. Structure of a mechatronic function module

Mechatronic function modules will be regarded in this paper in order to achieve a strong function orientation. The configuration of different MFM leads to an autonomous mechatronic system (AMS). A mechatronic product can be an AMS or a MFM.

2.2. Design of modular product platforms

A product platform can be defined as the collection of assets which are shared by products, including component designs [8]. This definition can be extended according to FELDHOUSE ET AL. who define a modular product platform as machines, subassemblies and components that can be combined as modules to fulfill different overall functions [9]. A module is further characterized by a low physical and functional dependency to other system elements and standardized interfaces that enable a reconfiguration [10]. The crucial step in platform design is the modularization. HÖLTTA-OTTO ET AL. provide an extensive overview of modularization methods [11]. Product platform design can be separated from the initial phase of platform planning and the subsequent phase of platform implementation [12]. SIMPSON ET AL. collected a broad overview of different approaches in these phases. SCHUH provides a holistic approach for the creation of modular product platforms integrating the planning, design and implementation in one process model [13]. All reviewed approaches lack a focus on function-oriented design and therefore have deficiencies when being applied to mechatronic systems. In this paper the focus is on a function-oriented design of modules for a mechatronic product platform.

2.3. Basics of axiomatic design

In this paper axiomatic design is identified as a suitable approach for the required function-oriented system representation and design. Hence the basics and application of axiomatic design will be explained briefly. Axiomatic design

(AD) was formulated by SUH in 1990 [14] and since then it has widely been applied to a variety of systems. The methodology offers a scientific base for the design of systems and improves design activities by providing the designer with a theoretical foundation [14]. The design process is understood as a consecutive mapping between four different domains:

- Customer domain – customer attributes (CA)
- Functional domain – functional requirements (FR)
- Physical domain – design parameters (DP)
- Process domain – process variables (PV)

Since this paper focuses on the design phase of a product platform the mapping of functional requirements and design parameters will be considered and further explained. FR and DP can mathematically be defined as vectors and their relationship can be represented by a design matrix [A]. The resulting mathematical representation is shown in the following equation.

$$\{FR\} = [A]\{DP\} \quad (1)$$

The Axiomatic Design methodology is based on two axioms which have to be respected: the Independence Axiom (1) and the Information Axiom (2). Axiom 1 states that an optimal design maintains the independence of FR and further that the adjustment of one DP satisfies a corresponding FR without affecting other functional requirements. According to AD a design can be *uncoupled*, *decoupled* or *coupled* depending on the design matrix resulting from the domain mapping. The three design characterizations are shown in the following equations. (X: influence; 0: no influence)

$$\text{Uncoupled:} \quad \begin{Bmatrix} FR1 \\ FR2 \end{Bmatrix} = \begin{bmatrix} X & 0 \\ 0 & X \end{bmatrix} \begin{Bmatrix} DP1 \\ DP2 \end{Bmatrix} \quad (2)$$

$$\text{Decoupled:} \quad \begin{Bmatrix} FR1 \\ FR2 \end{Bmatrix} = \begin{bmatrix} X & 0 \\ X & X \end{bmatrix} \begin{Bmatrix} DP1 \\ DP2 \end{Bmatrix} \quad (3)$$

$$\text{Coupled:} \quad \begin{Bmatrix} FR1 \\ FR2 \end{Bmatrix} = \begin{bmatrix} X & X \\ X & X \end{bmatrix} \begin{Bmatrix} DP1 \\ DP2 \end{Bmatrix} \quad (4)$$

For the modularization process in platform design the Independence Axiom is particularly important since modules require a functional independency. The optimal result of a modularization would be an uncoupled design with modules satisfying selected FR and not influencing other FR or DP.

Axiom 2 states that an optimal design is functionally uncoupled and has a minimum information content. Information content can be described as a “measure of the probability of success of achieving the specified FRs” [14]. In this context probability can be represented in a diagram using the design and the system range (see figure 2). The design range is determined by the selected DP. A functional requirement must be satisfied within the design range. The system range represents the possible range for acceptable outcomes. A FR fulfilment is given in the common range.

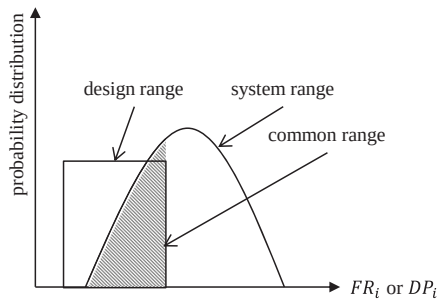


Fig. 2. Relationship between design range, system range and common range

In particular overengineering leads to high information content. Information content is directly related to complexity which should be reduced by modularization and platform design [15].

3. Methodology for the design of modular product platforms for mechatronic systems

The aim of the methodology is to support the design of modular product platforms for mechatronic systems. The initial point for the application of the presented method is a set of functional requirements (FR) which represents the platform requirements specification. FRs can be derived from customer attributes e.g. by applying the QFD method [14]. The outcome of the method is the definition of design parameters for a mechatronic product platform. Due to the high degree of functional integration and the resulting complexity of mechatronic systems a decomposition in mechatronic function modules (MFM) is proposed. A MFM as a subordinated system element can be designed individually to fulfil the assigned FR.

The methodology can be formalized as an extension of the axiomatic design approach. The corresponding graphical representation is illustrated in figure 3.

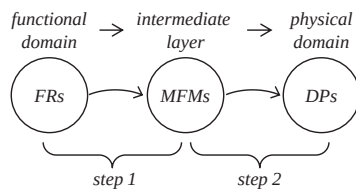


Fig. 3. Proposed extension of axiomatic design approach

In the following, the two steps of the methodology will be described in detail. In the first step, a proceeding for the decomposition of the system into mechatronic function modules is presented. In the second step, the design of MFMs for an application in a modular product platform is discussed.

3.1. Function-oriented modularization using an extended axiomatic design approach

The first step is a function-oriented modularization of the mechatronic system. The result are mechatronic function modules with delimited functional scope. To satisfy the

requirements of a modularization a low interdependency between the function modules is required. This will be achieved by applying an extended axiomatic design approach which includes the mapping of FRs to MFMs. The mathematical representation of this step is shown in the following formula:

$$\{FR\} = [A]\{MFM\} \quad (5)$$

At this point the FRs are known and it is necessary to identify corresponding MFM. Since a MFM is supposed to be a mechatronic subsystem which is complete and viable, it is necessary that all relevant subfunctions describing a mechatronic system are included [7]. To represent the functional scope required in a MFM a set of generic functions of a MFM was developed. The following four subfunctions can be derived from the main elements of a MFM:

- Information processing subfunction
- Actuator subfunction
- Mechanical subfunction
- Sensorial subfunction

Based on an analysis of the functional structure of components like actuators or sensors the subfunctions can further be broken down into partial functions. The result are eight generic functions which need to be included in the functional scope of a mechatronic function module (see figure 4).

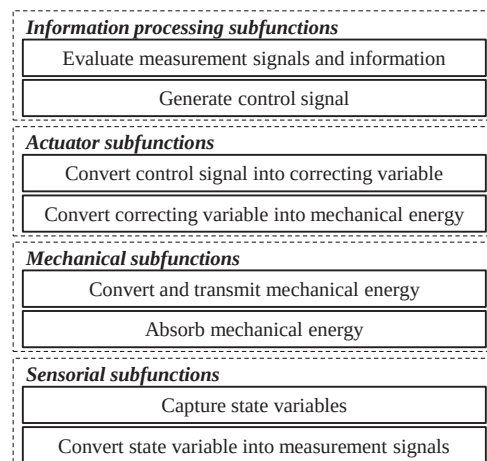


Fig. 4. Developed generic functions of a mechatronic function module

The derivation of MFMs from FRs can then be conducted using the developed generic functions (GF). If a FR can be split up into the eight generic functions it is suitable to be mapped to a mechatronic function module. This case is represented by the following equation:

$$FR_i = \begin{Bmatrix} GF1 \\ \dots \\ GF8 \end{Bmatrix} = MFM_i \quad (6)$$

It should be noted that this step already takes into consideration a certain knowledge of what technology is available. The process of technology selection for function modules is not addressed in this paper.

If the division results in less than eight generic functions the FR can be fulfilled for example only with mechanical subfunctions. One example would be the FR “transmit torque with 120 Nm”. This FR is likely to be fulfilled by a mechanical solution, in this case a gearbox. In order to correctly form a MFM it would be necessary to **increase the functional scope**. For instance, the FR “create torque” and “transmit torque” could be aggregated to one FR on the regarded level of the functional structure.

If the number of GF is higher than eight it is likely that the FR needs to be satisfied by more than one MFM or by a combination of a MFM and further architectural elements. An architectural element (AE) can for instance be a mechanical frame structure which only fulfills mechanical subfunctions. In this case it is proposed to **decrease the functional scope** which might result in a redefinition of the initial set of FRs.

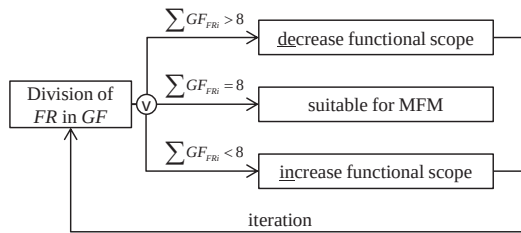


Fig. 5. Procedure for function-oriented modularization

Since the functional structure of a system can consist of different levels it can be necessary to apply an iterative proceeding comparable to the zigzagging described in axiomatic design [10].

The result will be a mapping of FR to MFM which satisfies the independence axiom (see chapter 2.3). A selected FR will be fulfilled by exactly one MFM. The functional dependence between MFM is assumed to be relatively low. The result of the mapping of FR to MFM is shown in an exemplary design matrix in equation 9.

$$\begin{Bmatrix} FR1 \\ FR2 \\ FR3 \\ FR4 \\ FR5 \end{Bmatrix} = \begin{bmatrix} X & 0 & 0 & 0 & 0 \\ 0 & X & 0 & 0 & 0 \\ 0 & 0 & X & 0 & 0 \\ 0 & 0 & 0 & X & (x) \\ 0 & 0 & 0 & (x) & X \end{bmatrix} \begin{Bmatrix} MFM1 \\ MFM2 \\ MFM3 \\ AE4 \\ AE5 \end{Bmatrix} \quad (7)$$

With the use of generic functions, a function-oriented modularization is achieved, which ideally satisfies the independence axiom. This is particularly favourable for platform design since it leads to a higher flexibility of the platform. It enables an easy substitution of function modules and the system independent design of MFM variants for use in a mechatronic product platform.

3.2. Design of a MFM for use in a product platform

In the second step, the MFM is designed for use in a product platform. Contrary to single product design in platform development ranges of FR need to be fulfilled. This can be clarified using the example of a machine tool. When

designing a machine tool for one specific purpose it could be demanded to fulfill a maximum spindle speed of 16.000 rpm for a specific purpose. The design of a product platform for machine tools could require the realization of maximum spindle speeds ranging from 16.000 to 40.000 rpm. For a product platform this results in a n-dimensional functional requirement space with “n” being the number of FR. This is illustrated in figure 6.

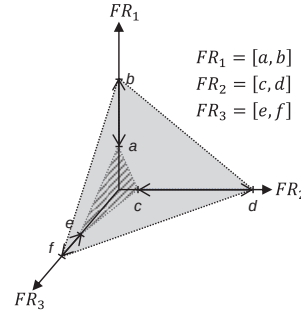


Fig. 6. 3-dimensional representation of a functional requirement space for a product platform

To fulfill the corresponding FR range it is necessary to identify suitable design parameters within the mechatronic function module. DP can be identified in the different disciplines of mechatronic systems (mechanics, electrics/electronics, software). Exemplary design parameters are different types of actuators or gearboxes. In order to identify all possible design parameters a mapping of generic functions to design parameters is conducted.

$$\begin{Bmatrix} GF1 \\ \dots \\ GF8 \end{Bmatrix} = \begin{bmatrix} X & \dots & 0 \\ \dots & \dots & \dots \\ 0 & \dots & X \end{bmatrix} \begin{Bmatrix} DP1 \\ \dots \\ DP8 \end{Bmatrix} \quad (8)$$

Since platform design requires FR ranges to be fulfilled it is necessary to identify which design parameters can be varied to fulfil FRs for different platform products. For the introduced design parameters this could be the type of actuator, the power of the actuator or the transmission ratio of a gearbox. The variation of a design parameter can always be conducted in a certain range which is determined by internal or external restrictions. A possible internal restriction is the capability of the company’s production facilities, while an external restriction could be the availability of technological solutions and suppliers. The feasible power of an actuator for a certain application for instance could range from 12 to 80 kW. The available range for the variation of a design parameter is termed the degree of freedom (DoF) of the DP (see equation 11).

$$DoF_i = [DP_{i\max} - DP_{i\min}] \quad (9)$$

Using the DoF of a certain DP it is possible to achieve a certain range of FR that could be fulfilled. The DoF is suitable for platform design if the variation results in a FR range that is at least larger than the FR range demanded for the product platform. Ideally, the DP variation would result in a FR range that exactly meets the demanded FR range and fulfils a defined minimum probability of function fulfilment in the boundary areas. This condition is illustrated in figure 7.

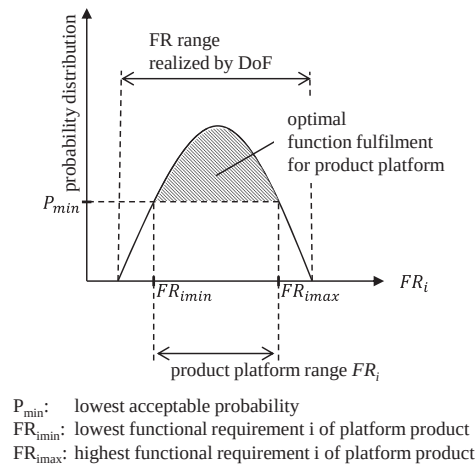


Fig. 7. Influence of design range on function fulfilment for a product platform

For the selection of favourable design parameters for the fulfilment of FR ranges the following criteria need to be regarded:

- Influence of DP variation (DoF) on fulfillment of FR
- Influence of DP variation (DoF) on additional cost
- Influence of DP variation (DoF) on other DPs

An ideal design realizes the FR range securely by the variation of only one DP which has a minimum influence on additional costs and no interdependencies to other DPs [16]. Mechatronic products are highly integrated systems and are usually characterized by high interdependencies between system parameters. Therefore it is likely that the variation of one DP results in the variation of further DPs. This change propagation leads to an increase of additional costs by primary and secondary change effects. These effects can be investigated by applying network analysis methods. Based on such analyses favourable design parameters for the fulfilment of the FR ranges can be selected. The application of the mentioned analysis methods is not further addressed in this paper.

4. Application in the development of a electromechanical steering system

The methodology was applied for the design of a modular product platform for electric vehicles. The steps introduced in chapter 3 are demonstrated for the preselected technology of electromechanical steering.

Prior to the application of the method the system and its functional requirements will be introduced. For the planned product platform three variants representing the full platform range will be considered. The main function “transport persons” can be broken down in the three functions “power vehicle”, “steer vehicle” and “decelerate vehicle”. Further functions are excluded from this case study for reasons of simplification. For these three functions performance indicators which are necessary for the quantification need to be defined. For the function “steering vehicle” the steering force in kN can be chosen. From the functional requirements

of the different variants the necessary range for this indicator can be identified constituting the functional requirement range for the product platform. A summary of all initial information is given in figure 7.

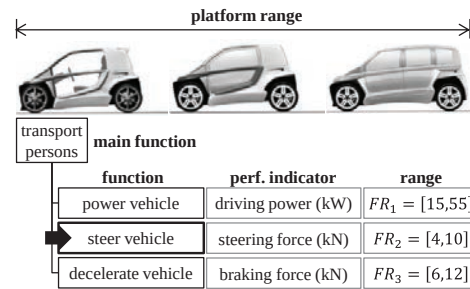


Fig. 8. Functions and FR ranges for electric vehicle platform

Step 1: Function-oriented modularization

Based on the introduced functional structure a first mapping of functional requirements to MFM is conducted. The result is a diagonal matrix which ideally satisfies the independence axiom (see equation 13).

$$\begin{Bmatrix} FR_1 \\ FR_2 \\ FR_3 \end{Bmatrix} = \begin{bmatrix} X & 0 & 0 \\ 0 & X & 0 \\ 0 & 0 & X \end{bmatrix} \begin{Bmatrix} MFM_1 \\ MFM_2 \\ MFM_3 \end{Bmatrix} \quad (10)$$

The denotation is given in the following table.

Table 1. FR and MFM for application case

Functional requirement	Mechatronic function module
FR_1 Power vehicle [15,55 kW]	MFM_1 Drive train
FR_2 Steer vehicle [4,10 kN]	MFM_2 Steering system
FR_3 Decelerate vehicle [6,12 kN]	MFM_3 Braking system

In the following, a closer investigation of the function “steer vehicle” will be conducted. To verify that FR_2 can be addressed by one MFM it is necessary to further decompose the function. Therefore a functional flow analysis of an electromechanical steering system is applied. The result is shown in figure 9.

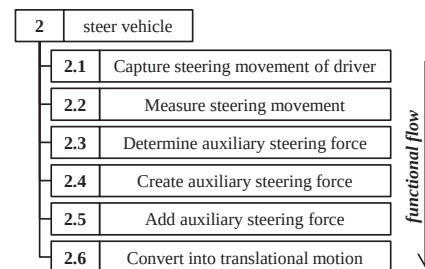


Fig. 9. Functional flow analysis of function “steer vehicle”

Since a MFM is required to contain all developed generic functions a mapping of the partial function resulting from the decomposition with the generic functions is conducted (see equation 11). It should be noted that a partial function can address more than one generic function (GF).

$$\begin{Bmatrix} FR2.1 \\ FR2.2 \\ FR2.3 \\ FR2.4 \\ FR2.5 \\ FR2.6 \end{Bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & X & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & X & X \\ X & X & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & X & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & X & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & X & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} GF1 \\ GF2 \\ GF3 \\ GF4 \\ GF5 \\ GF6 \\ GF7 \\ GF8 \end{Bmatrix} \quad (11)$$

The results shown above verify that the selected FR can be addressed by one specific MFM. This analysis has been conducted for further steering technologies. The selection of a superimposed electromechanical steering system for active steering or a steer-by-wire system would result in a system decomposition in two separate MFM. In this case a reformulation of the FR would be required. The clear delimitation of MFM is essential for the successful application of the subsequent step.

Step 2: Design of a MFM for use in a product platform

For the design of the resulting MFM “*electromechanical steering system*” a mapping of the generic functions to available design parameters is performed (see following design matrix).

$$\begin{Bmatrix} GF1 \\ GF2 \\ GF3 \\ GF4 \\ GF5 \\ GF6 \\ GF7 \\ GF8 \end{Bmatrix} = \begin{bmatrix} X & 0 & 0 & 0 & 0 & 0 \\ X & 0 & 0 & 0 & 0 & 0 \\ 0 & X & 0 & 0 & 0 & 0 \\ 0 & 0 & X & 0 & 0 & 0 \\ 0 & 0 & 0 & X & 0 & 0 \\ 0 & 0 & 0 & 0 & X & 0 \\ 0 & 0 & 0 & 0 & 0 & X \\ 0 & 0 & 0 & 0 & 0 & X \end{bmatrix} \begin{Bmatrix} DP1 \\ DP2 \\ DP3 \\ DP4 \\ DP5 \\ DP6 \end{Bmatrix} \quad (12)$$

The denotation is given in the following table.

Table 2. Generic functions and corresponding design parameters

Generic function	Design parameter
GF ₁ Evaluate measurement signals and information	DP ₁ Control unit
GF ₂ Generate control signal	DP ₁ Control unit
GF ₃ Convert control signal into correcting variable	DP ₂ Power electronics
GF ₄ Convert correcting variable into mechanical energy	DP ₃ Electric motor
GF ₅ Convert and transmit mechanical energy	DP ₄ Steering gearbox
GF ₆ Absorb mechanical energy	DP ₅ Housing
GF ₇ Capture state variables	DP ₆ Steering torque sensor
GF ₈ Convert state variable into measurement signals	DP ₆ Steering torque sensor

Information processing and sensorial subfunctions are realized by one design parameter and not split according to their generic functions. An uncoupled and square matrix would represent the highest possible modularity. In the following it is necessary to identify favourable design parameters for the fulfilment of the FR range (4-10 kN steering force). For each DP possible degrees of freedom

(DoF) were derived. Through a network analysis which takes into consideration the three criteria introduced in chapter 3.2 an investigation on all derived DoF was performed. The analysis resulted in the recommendation to realize the FR range by adapting the transmission ratio of the gearbox and the windings of the electric motor. These parameters need to be flexible for the derivation of single products from the platform. All other DoF need to be specified to one value and kept invariant for all future platform products.

5. Conclusion

The introduced methodology enables a function-oriented design of modular product platforms for mechatronic systems. In a first step the mechatronic system is modularized in mechatronic function modules using generic functions in the mapping process. In the second step MFM are designed for the use in a product platform. The generic functions are mapped to design parameters that enable the fulfilment of the functional requirement. Since a product platform needs to fulfill ranges of FR the main task is to select favorable design parameters for the necessary variation. These selected design parameters are responsible for the integration of the necessary flexibility in the product platform. The methodology was applied to an electromechanical steering system proofing the applicability. Further research will focus on change propagation effects resulting from design parameter variation.

References

- [1] Hehenberger P. Perspectives on hierarchical modeling in mechatronic design. *Advanced Engineering Informatics* 28; 2014. p.188-197.
- [2] Buur J.: A theoretical approach to mechatronics design. Dissertation. Technical University of Denmark, 1990.
- [3] Elmaraghy H, Schuh G, Elmaraghy W, Piller F, Schönsleben P. Product variety management. *CIRP Annals*. 62(2); 2013. 629-652.
- [4] Schuh G, Amoscht J, Rudolf S. Integrated Development of Modular Product Platforms. In: *Proceedings of PICMET '10*. Phuket, Thailand, 18-22 July 2010, pp. 1928-1940.
- [5] Schuh G, Rudolf S, Schiffer M, Schrey E. *Modular Product Platform and Variant Management 2014*. Conclusions of the Consortial Benchmarking. Aachen, Germany; 2015.
- [6] VDI-Richtlinie 2206, *Entwicklungsmethodik für mechatronische Systeme*; 2004.
- [7] Lueckel J, Hestermeyer T, Liu-Henke X. Generalization of the Cascade Principle in View of a Structured Form of Mechatronic Systems. *Proceedings of the IEEE/ASME International Conference on Advanced Intelligent Mechatronics*; Como, Italy, 8-12 July 2001.
- [8] Ulrich KT, Eppinger SD, *Product design and development*; 2007. p. 180.
- [9] Feldhusen J, Grote KH; Pahl/Beitz *Konstruktionslehre*; 2013. p. 839.
- [10] Göpfert J, *Modulare Produktentwicklung*. Dissertation, LMU Munich, 1998.
- [11] Hölta-Otto K, Otto KN, Simpson TW. Defining Modules for Platforms: an overview of the Architecting Process. in Simpson et al.. *Advances in Product Family and Product Platform Design: Methods & Applications*; Springer; New York; 2014. p. 323-341.
- [12] Simpson T, Jiao J, Siddique Z, Hölta-Otto K. *Advances in Product Family and Product Platform Design: Methods & Applications*; Springer; New York; 2014.
- [13] Schuh G. *Leitfaden zur Baukastengestaltung*; VDMA; Stuttgart; 2015.
- [14] Suh NP. *The principles of design*. Oxford University Press; 1990.
- [15] Suh NP. Complexity in Engineering. *CIRP Annals-Manufacturing Technology* 54 (2); 2005. p. 46-63.
- [16] Schuh G, Rudolf S, Amoscht J, Lüdtk B. Increasing commonalities by designing production-oriented modular product platforms. *Advanced Materials Research*. Vol.907; 2014. p. 197-210.